

Toward Inclusion Beyond Language: Proposing the *Polylingual Co-Design Method*

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We introduce Polylingual Co-Design, a participatory design method enabling cross-cultural collaboration without shared language. Participants communicate exclusively in their native tongues while engaging in collaborative design activities, preserving natural paralinguistic elements like intonation and gesture. Our pilot study with Finnish and Telugu speakers designing a guesthouse party demonstrates that non-verbal communication through drawings, expressions, and vocal modulation can facilitate effective co-creation. The method surfaces language-independent design concepts while maintaining cultural authenticity, offering new possibilities for inclusive participatory design in multilingual contexts.

1 Introduction

The HCI community today recognizes that systems can no longer be designed in standardized American English alone[1, 3, 6]. Structural changes such as global migration, the internationalization of digital platforms, and the expansion of remote collaboration have made it more common for design processes to involve people who speak different first languages and may not share a fully fluent common language. This growing linguistic diversity foregrounds the issue that design artifacts forged within a single linguistic worldview risk exclusion or misinterpretation when encountered by users who do not share that language[4]. Prior studies have shown that language shapes how people interpret social norms and metaphors, making it difficult to preserve intended meaning across linguistic boundaries—even with careful translation[4, 6]. A recent example is Danggeun-Market’s “Manner Temperature” system, which visualizes a user’s reputation as a body temperature—implying that more considerate users are “warmer.” While warmth in Korean culture connotes kindness and trust, international users expressed confusion: some wondered why a “cool” person should be seen less positively[7]. This situation underlines the need for **language-independent design**—design whose core meaning can be understood without relying on any single linguistic code.

Yet, despite repeated calls for inclusivity, responses have largely focused on translation and localization—treating language as a downstream variable to manage after core design decisions are made[1]. Some efforts have sought to engage with linguistic diversity, through the inclusion of multilingual participants in co-design. These methods typically rely on two strategies: (1) introducing technological mediation such as real-time captioning[3] or speech-to-speech translation[2], or (2) eliminating verbal language altogether, using pantomime[5] or silent prototyping[9] to avoid linguistic asymmetry. Although these approaches expand access and participation, they introduce trade-offs. Both tend to sacrifice the rich paralinguistic features of natural speech—intonation, rhythm, turn-taking—that play a vital role in co-creative dialogue and negotiation. When language is either filtered through machines or completely removed, subtle cues that support mutual adjustment, emotional resonance, and creative friction are diminished[2, 9]. These limitations point to an underexplored methodological possibility: maintaining

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the expressive and interactional qualities of natural speech, while removing the cultural and semantic assumptions embedded in any particular language.

There, we introduce ***Polylingual Co-Design***—a new design methodology in which the setup requires participants to speak in their mother tongues, despite the absence of mutual intelligibility. We hypothesize that this setup can surface language-independent elements of design that are comprehensible across linguistic boundaries. To explore its feasibility and value, we conducted a pilot study in which participants from different linguistic backgrounds were tasked with co-designing a guesthouse party. This paper makes the following contributions:

- (1) **Conceptual framework of *Polylingual Co-Design*** – we propose a novel methodology for exploring language inclusivity in design, and provide detailed documentation of process, materials, and facilitation techniques to enable replication.
- (2) **Empirical insights from a pilot study** – we present qualitative analysis of how participants communicated and co-created in a polylingual setting, identifying communication patterns, challenges, and nature of design outcomes.
- (3) **Implications for language-independent design** – we reflect on how our findings can inform participatory design practices seeking to surface design concepts that transcend specific linguistic contexts.

2 Design Situation and Motivation

As multilingual contexts become more prevalent, language inclusivity has become a foundational consideration within the HCI community. While linguistic diversity is addressed across most contexts, this study focuses on situations where establishing a shared language is not feasible. When users’ linguistic backgrounds are diverse, relying on a shared language can marginalize certain groups. There, conventional co-design practices that presuppose linguistic homogeneity face limitations, creating a need for alternative methodologies that do not privilege any single language. In addition, we focus our scope to multilingual contexts where design decisions require negotiation and consensus among users. Existing approaches that eliminate verbal language altogether or rely on translation technologies tend to suppress non-verbal cues, reducing the dynamics of interaction. These limitations can constrain the interactional flexibility of co-design, which is critical for co-ideation and mutual adjustment. In such contexts, an inclusive methodology that preserves the expressive qualities of speech can serve as a valid alternative. The following design situations illustrate contexts where linguistic complexity and negotiation-driven interaction are particularly salient.

- **Community-based public design in multicultural settings** - In multicultural communities, linguistic diversity is an everyday reality, making it difficult to designate a shared language. Democratic participation is essential for inclusive decision-making, yet communication through a dominant language risks excluding certain groups. In such contexts, there is a need for a methodology that do not assume a shared language, yet still enable meaningful interaction among users, allowing creative friction to emerge.
- **Public services used by linguistically diverse and transient populations** - In contexts such as airports or tourist sites, public services are often used simultaneously by large numbers of people of different linguistic backgrounds. Because the linguistic profile of users is both unpredictable, it is not feasible to define a default language in advance. Translation-based solutions are also limited in their capacity to cover all possible languages in real time. These environments call for participatory approaches that reduce language dependency and promote universally interpretable designs.
- **Early-stage concept development for global products** - Early concept development often faces cultural misalignment when language-specific meanings don’t transfer across contexts. Approaches that help

uncover language-independent concepts are needed. In particular, co-design methods that allow participants to negotiate meaning without relying on shared language can support more inclusive and globally relevant outcomes.

3 *Polylingual Co-Design* Method

3.1 Implementation and Procedure

The *Polylingual Co-Design* method is structured around a core principle: participants communicate exclusively in their native languages while engaging in collaborative design activities, without any form of linguistic mediation or translation. This principle in the method preserves the natural paralinguistic elements of speech—prosody, intonation, gesture, and rhythm. The method follows a structured three-phase approach: session briefing and ice-breaking, polylingual co-design activity, and post-interview reflection.

Phase 1 – Session Briefing and Ice-Breaking. : The session is opened with an overview given in common language or with interpretation. This includes a description of the task, an explanation of the language limitation that requires participants to use only their native languages, and note that the exercise is open-ended. Participants then introduce themselves in their respective native tongues. Despite not being able to understand each other verbally, this activity helps individuals to establish vocal familiarity and become familiar with the method.

Phase 2 – Polylingual Co-Design Activity. : The co-designing task can vary by each design goal and can be divided into multiple stages, carried out by the moderator. The moderator maintains a supportive but non-interventionist stance, upholding linguistic constraints while providing minimal structural guidance—ensuring smooth transitions between phases, managing time, and intervening only when communication completely breaks down. Crucially, facilitators should not translate, interpret, or bridge semantic gaps, instead allowing participants to develop their own collaborative rhythms and cross-cultural communication strategies. Multiple facilitators as observers should be present to capture the complexity of non-verbal interactions and cultural nuances that single-person moderation cannot adequately document.

Phase 3 – Post-Interview Reflection. : The participants engage in a group interview, conducted in the common language or with interpretation. The goal is to collect reflections on the session, clarify perceived meanings, and uncover insights or ideas that may not have arisen during the co-design process itself.

3.2 User Involvement in the Method

Level of User Involvement. While some research emphasizes the importance of micro-dynamics in participatory interactions[8], our method requires active involvement of users to ensure better co-design outcome. Given the experimental nature of our method and the inherent communication barriers, participants will naturally maintain heightened attention. However, still the facilitator guidance becomes crucial to sustain this involvement and focus, to encourage mutual understanding and creative momentum.

User Activities and Tasks. Participants engage in flexible design tasks according to different design goal of the researcher. User activities are towards creation and divergent thinking through generative activities, while post-interview session is also about reflection of the session communication itself.

Data Collection. The method generates two primary data types: tangible artifacts and interpretive data. Tangible artifacts include common design result and individual communication sketches. Interpretive data includes post-interview reflections as well as observational data such as notes taken by the observer (researcher) and the video or voice recording taken under consent.

Data Interpretation. Interpretation approaches vary based on design needs and contexts. Designers can analyze elements such as design features accepted by all participants despite language barriers and may identify language-transcendent design concepts. Post-interview insights may reveal where cultural assumptions affected understanding, indicating opportunities for language-independent design development. Multiple sessions across different participant combinations can help distinguish generalizable patterns.

3.3 Prototyping Materials

Even without a shared language, natural communication methods such as gestures, facial expressions, and vocal intonation can provide support for interaction. However, to enhance cross-cultural communication effectiveness, our method allows participants to utilize visual expression through drawing materials. Drawing serves dual purposes: it provides visual expression that transcends linguistic barriers and enables collaborative creation through shared visual language. Therefore, our prototyping approach provides drawing materials alongside additional interaction catalyst tools to facilitate further collaboration support, with the primary goal of enabling deeper participant engagement and more effective

Visual Materials: Paper and Drawing Tools. The method employs simple, accessible materials centered around paper-based drawing systems. We provide two distinct spatial arrangements to support different communication and collaboration needs. Individual spaces consist of personal drawing sheets where participants can develop initial ideas in their preferred visual style and pace, also used for direct communication with the other participant. Shared spaces feature large collaborative canvases where participants combine and build upon mutual agreement, facilitating collective design results. Basic drawing tools selection of pen, markers, and crayons prioritizes accessibility to ensure broad participation—paper and basic drawing tools require no specialized skills or cultural knowledge, allowing immediate engagement from all participants regardless of design experience. The material system can be expanded to include tactile elements such as clay for three-dimensional concept exploration, providing additional modalities for design tasks involving spatial or sculptural expression.

Collaboration Support Tools: Interaction Catalyst. Beyond basic drawing materials, the method incorporates interaction catalyst tools designed to facilitate structured collaboration. These tools should be serving as neutral facilitators rather than culturally-specific implements. Our pilot study employed flag and ticket systems as collaboration support tools, which will be detailed in the pilot study section.

4 Pilot Study and Results

To evaluate the potential of *Polylingual Co-Design* as an inclusive approach, we carried out a pilot study that reimagines the importance of verbal language in participatory activities. Participants were instructed to communicate solely in their native languages, even if they couldn't understand each other directly.

4.1 Study Setup and Procedure

Participants. The pilot study involved two female students with entirely different native languages: Finnish and Telugu (Table 1). Finnish and Telugu were deliberately paired as linguistically and culturally distant, minimizing

Table 1. Participant Demographics

P	Age	Gender	Languages Known	Mother Tongue (Used)	Academic Background
P1	24	F	Finnish, English, Swedish, Italian	Finnish	Nuclear and Quantum Engineering
P2	20	F	Telugu, Hindi, English, Mandarin	Telugu	Business Administration

accidental understanding. Participants never dealt with design, a deliberate choice to explore how non-designers navigate both linguistic and methodological constraints in collaborative tasks.

Design Task. The co-design task asked participants to imagine a fictional party taking place in a guesthouse, chosen deliberately for its naturally international atmosphere. Guesthouses often gather people from diverse cultural and linguistic backgrounds, frequently without the assurance of shared language comprehension. This made it an ideal setting to examine the practical application of *Polylingual Co-Design* in real-world scenarios. The brief did not require any formal output or stylistic uniformity. Instead, the emphasis was on examining communicative strategies, non-verbal negotiation, and the design decisions made by participants to create a shared vision.

Prototyping Materials. Blank papers are given for expression—one in the center as a shared space for co-creation and the others prepared for each participant, working as individual spaces. Drawing tools including pens and markers are provided together. Two additional tools are introduced to support interactions. (1) Free Translation Ticket: The chance for requesting the translation of a single word, created as emergency tools to promote reliance on non-verbal strategies. (2) Status Flags: Pair of small flags which are green—for “*I understand*”—and white—for “*I don’t understand*,” which are used to signal the status of comprehension without interrupting the interaction.

Researcher Roles. Our pilot study employed a four-person research team with distributed moderation responsibilities. One researcher served as the primary moderator, introducing the design task and guiding session transitions, while three others functioned as silent facilitators who supported workflow without verbal intervention. All team members simultaneously observed and documented non-verbal interactions, material usage patterns, and spatial dynamics through notes, video recording, and behavioral mapping. During the concluding group interview, two researchers led the discussion while the remaining two provided targeted follow-up questions, creating a comprehensive understanding of both the collaborative design experience and participants’ methodological feedback.

Procedure Steps. The pilot study was carried out as the following:

Phase 1 – Session Overview and Ice-Breaking (10 minutes): Session is briefed in English (common language) with visual aid slides. Participants were introduced to the design task explained above and started by introducing themselves in native language.

Phase 2 – Polylingual Co-Design Activity (24 minutes): To facilitate the process and prevent disorientation, the session was segmented into three thematic stages. Each stage lasted for 8 minutes and had a concentrated design focus as the following:

- (1) **Stage 1 - Table and Food:** Imagining the table arrangement and chose the food, inspired by cultural connections to sharing and togetherness.
- (2) **Stage 2 - Space and Atmosphere:** Focusing on the spatial arrangement, lighting, decor, and overall ambiance of the party.

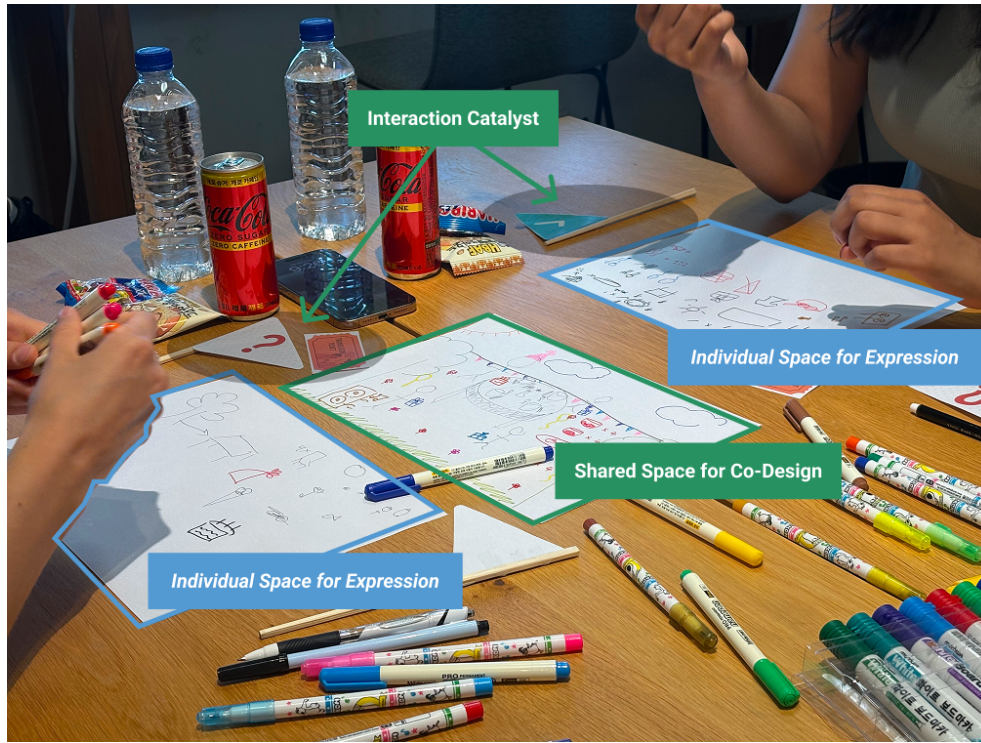


Fig. 1. Setup of Pilot Study

- (3) **Stage 3 - Activities and Events:** Suggesting organizing entertainment, games, or group rituals to occur during the event.

Phase 3 – Post-Interview (15 Minutes): The participants engaged in a brief group interview, conducted in English. Participants were encouraged to share their reflections on the session, clarify communication they had earlier and identify interesting moments during the co-design process.

4.2 Findings and Insights

The analysis of data collected during the pilot study, including observations, visual artifacts, and participant interviews, revealed key reflections on the effectiveness and limitations of the *Polylingual Co-Design* method.

4.2.1 Communication without Shared Language.

Non-verbal Communication. In the absence of a shared language, communication unfolded through drawings, gestures, vocalizations, and gaze. The inability to articulate ideas verbally enhanced the use of visual and physical strategies: participants drew, pointed, mimed, and modulated their tone of voice to convey intentions and reach consensus. Participants generally reported that they understood each other, even when speaking in different languages. In many cases, they believed they had understood their partner even when misunderstandings or misinterpretations had actually occurred. Participants often mentioned that they could extract meaning from each other's facial expressions and gestures, and it helped them to feel like understanding each other.

Interaction Tools. Despite the availability of Status Flags and Free Translation Tickets, neither participant felt the need to use them. The spontaneous system of communication, built through drawing, gesture, and vocal modulation, proved sufficient for completing the co-design task. This behavior suggests that, within a supportive setting, explicit support tools may become redundant, making room for more fluid and intuitive forms of expression.

Partial Comprehension through Language. In some cases, participants were able to recognize and understand specific words despite the language barrier. This occurred through various mechanisms: lexical similarities between the two languages, sound symbolism effects such as the bouba-kiki phenomenon, or partial mutual intelligibility (e.g., similarities between Finnish and English). Compared to drawing or gesturing, recognizing a spoken word enabled faster and more direct communication, often accelerating the flow of interaction.

4.2.2 Collaborative Strategies.

Offer-and-Choose Pattern. Rather than progressing through top-down reasoning or mutual deliberation, collaboration typically unfolded in a pattern where one participant proposed an idea and the other either accepted or responded. Visual aids such as sketches were frequently used to offer options, which the other participant was then invited to select.

Accepting over Arguing. Throughout the decision-making process, participants tended to accept rather than resist each other's proposals. As long as mutual comprehension was achieved and the idea did not raise major objections, it was usually added to the co-design board without challenge. Only two instances of explicit rejection were observed; in both, the proposing participant quickly abandoned the idea without attempting to persuade the other. This suggests that in polylingual settings, active negotiation is minimal, and participants are more likely to identify divergence points passively.

4.2.3 Co-creation and Resulting Artifacts.

Language-independent Design. Participants spontaneously oriented their designs toward accessible and culturally neutral elements (e.g., balloons, flowers, music), which supported mutual understanding but often led to generic outcomes lacking in cultural specificity. This convergence within the "commons" highlights the inclusive potential of the method, while also revealing a risk of symbolic flattening.

Shared Outcomes vs. Individual Ideas. Culturally specific ideas were sometimes left unshared during the session but emerged later through personal sketches or post-session interviews. This suggests that while consensus was achieved, the method may induce a form of self-censorship for concepts that are difficult to express non-verbally. A triangulated analysis of observations, artifacts, and interviews provided a fuller picture of the experience.

5 Discussion

5.1 Implications and Method Guidelines

Our findings offer valuable implications for participatory design, particularly in cross-cultural and polylingual settings. The observation on non-verbal cues underscores the potential of subtle, multi-sensory communication elements in facilitating mutual understanding even without shared verbal language. Based on these insights, we propose several design considerations and guidelines to inform future participatory design practices in polylingual contexts.

Participant recruitment should prioritize diversity and psychological safety. It is crucial to intentionally recruit participants from a wide range of linguistic and cultural backgrounds to ensure diversity of perspectives. Equally important is creating an environment in which individuals feel comfortable expressing themselves in their native language without pressure to conform to a common spoken language. Psychological safety, fostered through the acceptance of multiple modes of expression, enables more authentic and equitable participation.

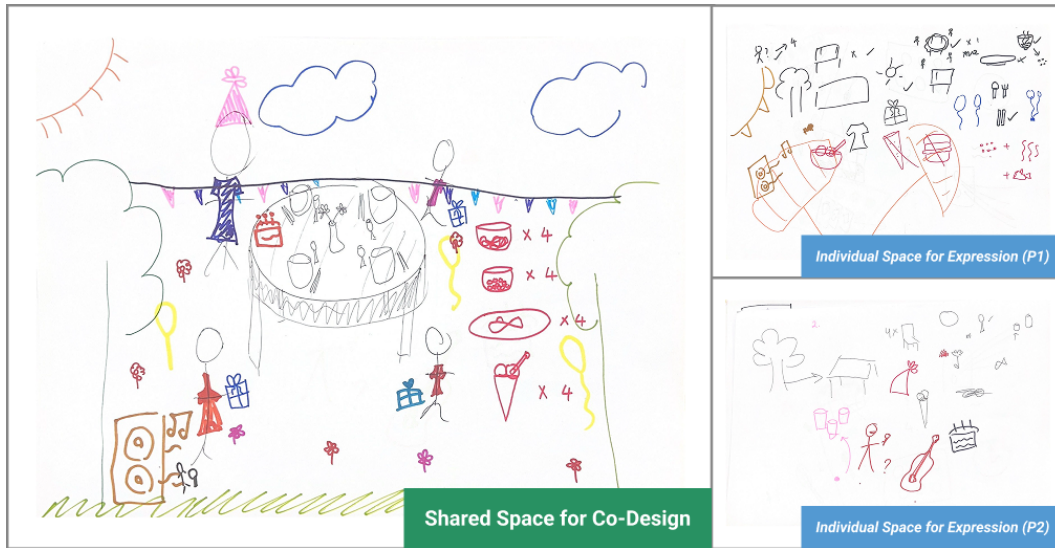


Fig. 2. Co-created Artifacts as Outcomes

Design tasks and materials should be intentionally crafted to promote easy ideation. Tasks must be designed to encourage visual and gestural communication from the outset, ensuring that all participants, regardless of linguistic background, can engage meaningfully. Materials used in these activities should be intuitive, and culturally neutral. Tools such as colored paper, thick markers, crayons, and simple craft supplies like clay support a wide range of expressive possibilities without requiring specialized skills. Moreover, the expressiveness of the materials should be matched to the demands of the specific design task to ensure that key nuances can be adequately conveyed.

Tools based on the polylingual setting must be considered if necessary. While basic materials may suffice, any additional tools introduced should be critically evaluated to determine whether they genuinely enhance communication or inadvertently create barriers. In some cases, the deliberate absence of complex tools can support more inclusive dialogue.

Finally, during the **interpretation** stage, careful attention must be given to how insights are derived and generalized. Researchers must assess whether the design tasks and outcomes exhibit a degree of neutrality or universality, enabling their use across different cultural contexts. If such universality is not evident, post-evaluation efforts should focus on refining or adapting the outputs to ensure broader applicability. This interpretive process is essential to maintain the inclusivity of participatory design outcomes in diverse, polylingual settings.

5.2 Limitation and Future Work

While our method offers promising directions for cross-cultural participatory design, several limitations remain. First, inconsistencies could be emerged in the design process due to differences in participants' drawing skills, personalities, and cultural backgrounds. These variations affected the clarity of communication and could differ participants in visual expression. Furthermore, while the method effectively supported low-level ideation, it encounters difficulties in addressing tasks that required complex abstraction or technical specificity.

Another limitation is the tendency toward over-generalization. The need for universally understandable communication sometimes led to safe, common ideas rather than more specific or innovative solutions that are harder to express without language. For the future work, to address limitations related to inconsistencies in the

methodology, a robust approach involves conducting multiple pilot studies. This allows for iterative refinement and the identification of potential issues.

6 Reflection

Creating the pilot study for the *Polylingual Co-Design* method was deeply formative, both methodologically and personally. While defining a new method was challenging, our unique setup made the process feel natural. Seeing how meaning emerged through drawings, gestures, and voice shifts challenged the idea that spoken language must always be central in co-design, showing us the power of alternative forms of expression. Looking more closely at the work done, one of the most revealing aspects was realizing that none of the interaction tools we had prepared (tickets and flags) were actually used. This was not oversight but spontaneous choice: participants built an alternative shared language that was more fluid and natural. This dynamic offered important insight: in certain contexts, minimal mediated interaction can be more effective than overly structured setup. It is the classic case where designers develop expectations around method usage, to discover user behavior often diverges from expectations, even when the proposed solution seems valid. Another striking aspect was the delicate balance between shared understanding and cultural depth. The ideas that emerged during the session were undoubtedly inclusive and accessible, but many personal insights or culturally rich references remained implicit, only resurfacing in the individual materials or in the post-session discussions. This raises an important question: To what extent should inclusive design aim for convergence? When should it allow space for divergence instead?

Moreover, this methodology grew from the real struggles we faced as a team, which made it feel deeply personal. We often used part of the method unconsciously, even when sharing a language. Realizing that the pilot echoed our own experience made it even more meaningful, and reflecting on how that could become a method was truly stimulating. Ultimately, this project was not just about testing a method, it challenged us to engage in new forms of listening, observation, and understanding, which are fundamental skills for any designer who is curious, aware, and willing to empathize with others. In this sense, it expanded not only what co-design can be but also how we, as researchers, choose to live and experience it.

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